HyPyraMamba: A Pyramid Spectral Attention and Mamba-Based Architecture for Robust Hyperspectral Image Classification

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***Abstract*—In hyperspectral image (HSI) classification, the high-dimensionality and the complex coupling of spatial–spectral features present severe challenges to existing deep learning methods in terms of accuracy, generalization, and computational efficiency. Researchers have recently explored convolutional neural network (CNN) and Transformer-based methods to overcome these limitations, but CNN's limited receptive field prevents effective modeling of long-range dependencies, while Transformers suffer from high computational cost and inefficiency in high-dimensional data. Motivated by these limitations, the state-space model (SSM) Mamba shows great potential as an efficient alternative for sequence and dependence modeling. Building on this foundation, we propose HyPyraMamba, a novel architecture designed to effectively overcome the above challenges. It integrates the pyramid spectral attention (PSA) module to capture multiscale key spectral features, thereby reducing interference caused by spectral redundancy. We developed an adaptive expert depthwise convolution (AEDC) module that enhances the model's ability to express multiscale spatial–spectral features, and a sequence modeling module, Mamba. In the Mamba module, we utilize the spatial Mamba and spectral Mamba branches to enhance spatial structure and spectral correlation modeling. Extensive experiments on four benchmark HSI datasets demonstrate that HyPyraMamba significantly outperforms several recent state-of-the-art methods and provides a favorable accuracy–efficiency tradeoff. In particular, classwise analyses on spectrally similar land-cover categories (e.g., different soybean and bareland types) show that HyPyraMamba markedly reduces mutual confusion compared with CNN-, Transformer-, and Mamba-based baselines. The code will be available at https://github.com/dekai-li/HyPyraMamba**

***Index Terms*—Hyperspectral image (HSI) classification, Mamba, pyramid spectral attention (PSA), spatial–spectral fusion.**

Received 22 October 2025; revised 4 December 2025; accepted 29 December 2025. Date of publication 1 January 2026; date of current version 9 January 2026. This work was supported in part by the National Science Foundation of China (NSFC) under Grant 62463006; in part by Hainan Province 2025 International Science and Technology Cooperation Research and Development Project under Grant GHYF20250; and in part by the Scientific and Technological Research Program of Chongqing Municipal Education Commission under Grant KJZD-K202400604. *(Corresponding author: Uzair Aslam Bhatti.)*

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Digital Object Identifier 10.1109/TGRS.2025.3650350

I. Introduction

HYPERSPECTRAL images (HSIs) hold a significant position in remote sensing, as they not only contain rich spatial information but also include hundreds of continuous spectral bands, enabling the detailed representation of the spectral characteristics of objects [1], [2], [3], [4]. However, the high dimensionality, information redundancy, and spectral confusion inherent in hyperspectral data pose significant challenges for accurate classification. Traditional classification methods, such as support vector machines (SVMs) [5], linear discriminant analysis (LDA) [6], and manifold learning-based methods [7], typically rely on manually designed features or subspace dimensionality reduction techniques, which are often ineffective in capturing the complex spectral–spatial structure of hyperspectral data.

In recent years, the rapid development of deep learning has significantly advanced HSI technology [8], [9]. Methods based on convolutional neural networks (CNNs), for example, effectively extract local spectral–spatial information from hyperspectral data through the properties of local receptive fields and weight sharing [10]. For instance, the spectral–spatial residual network (SSRN) based on 3-D convolution, by utilizing a joint feature extraction mechanism, has effectively improved the classification accuracy of HSIs [11]. However, convolutional operations are typically limited to local feature perception and struggle to capture global spectral–spatial contextual dependencies, thereby restricting further improvements in model classification performance.

In contrast, Transformer models, centered on the self-attention mechanism, have shown significant advantages in HSI classification tasks due to their excellent ability to model long-range dependencies [12], [13]. Existing studies have developed models such as HSI-BERT [14] and SpectralFormer [15], which effectively capture global information in the spectral or spatial dimensions. However, the self-attention mechanism in Transformer models inherently has quadratic computational complexity [$O(N^2)$], which leads to lower computational efficiency and higher memory requirements when processing high-dimensional data, thereby limiting

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their widespread application in high-dimensional hyperspectral scenarios.

The recently emerging Mamba network based on state-space models (SSMs) has linear computational complexity, effectively overcoming the bottleneck of Transformer methods and demonstrating exceptional performance in processing long sequence data [16]. However, traditional Mamba models are typically limited to 1-D or 2-D data and exhibit significant shortcomings when handling 3-D HSI data. Moreover, existing Mamba models have limited selective scanning capability in the spatial–spectral dimensions, making it difficult to fully exploit the spatial–spectral information of hyperspectral data [17]. This limitation motivates us to explore a 3-D Mamba network structure suitable for high-dimensional hyperspectral data, capturing global spectral–spatial dependencies and further improving the model's classification performance and computational efficiency. In summary, the field of HSI classification currently faces the following key challenges that need to be addressed:

1) the issue of severe feature redundancy in high-dimensional spectral data and how to effectively extract key spectral information;
2) the complex coupling of spatial–spectral features and how to fully and effectively integrate spatial and spectral information;
3) the issue of high computational complexity in existing long-sequence modeling methods (such as Transformer) which makes it difficult to efficiently apply them to high-dimensional data.

To address the issues, this article proposes a novel 3-D pyramid spectral–spatial attention Mamba network. First, the model introduces the pyramid spectral attention (PSA) mechanism, which effectively and adaptively captures multiscale key spectral features, thereby reducing interference caused by spectral redundancy. Next, the design of two submodules—spatial Mamba and spectral Mamba—enables the extraction of spatial and spectral features, achieving effective fusion and complementarity of spatial–spectral information. Additionally, the incorporation of the adaptive expert depthwise convolution (AEDC) module enhances the model's ability to express multiscale spatial–spectral features, improving both generalization performance and robustness. Fig. 1 highlights the research gap and innovation motivation for developing the proposed model. The main contributions of this article are as follows.

To the best of authors' knowledge, the proposed HyPyraMambaNet is the first spatial–spectral network combining the PSA mechanism with spatial and spectral Mamba networks, followed by the AEDC module to achieve efficient and accurate joint modeling of spatial–spectral features.

An efficient PSA module is designed, capable of adaptively capturing multiscale spectral–spatial feature information, significantly enhancing the effectiveness of feature representation, which solves other attention mechanisms of spectral redundancy by emphasizing informative wavelengths.

To tackle the limitations of CNN, Transformers, and Mamba networks, we introduced the AEDC module after the spatial–spectral Mamba network architecture for the first time,

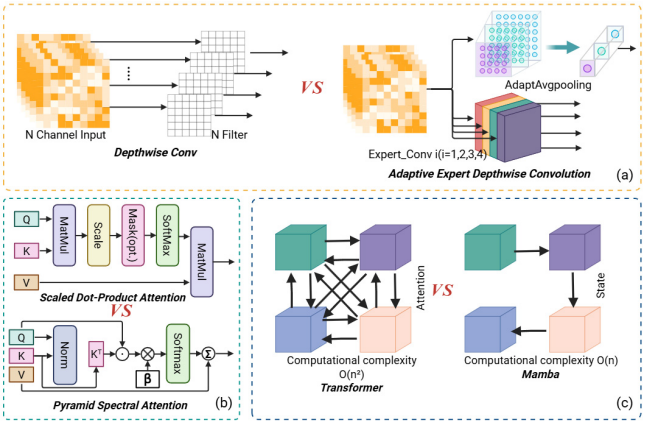


Fig. 1. Comparative analysis of innovation proposed in this study. (a) Depthwise versus AEDC module. (b) Scaled attention versus PSA module. (c) Transformer versus Mamba.

which further improves the model's ability to perceive complex spatial–spectral information and enhances classification accuracy. AEDC links multiple expert kernels with different receptive fields to get varying spatial–spectral information, while other convolution networks struggle to achieve this information.

We performed extensive experimental validation on four complex hyperspectral datasets—Indian Pines (IP), LongKou, Salinas, and Xuzhou. Comparison with the last three years' latest HSI classification methods demonstrates that the proposed model significantly outperforms existing methods in terms of accuracy, robustness, and generalization ability.

The remainder of this article is organized as follows. Section II reviews related work, Section III introduces the basic concepts and the proposed method, Section IV presents the experimental analysis, and Section V concludes this article.

II. Related Work

Traditional machine learning methods have made significant contributions to HSI classification tasks. For example, Okwuashi and Ndehedehe [18] proposed a deep SVM model for HSI classification, implemented using four kernel functions: exponential radial basis function (ERBF), Gaussian radial basis function (GRBF), neural networks, and polynomials. Independent support vectors constitute the interconnection weights of the entire network. The network is trained using a dataset of 100 input samples, and the model's regularization parameters are used to initialize the interconnection weights. Huang et al. [19] leveraged the nonlocal principle of real images through K-nearest neighbors (KNNs), achieving competitive classification results and fast computational efficiency.

With the continuous rise of deep learning, more solutions have been proposed for HSI classification tasks. Hsieh and Kiang [20] explored a 1-D CNN (1D-CNN) based on pixel-level spectral data, using 1D-CNN for specific spectral bands and spectral–spatial features for agricultural HSI classification, achieving good classification performance. Traditional methods struggle to simultaneously leverage spatial context and spectral correlations. Therefore, Yang et al. [21] proposed a 2-D CNN (2D-CNN) model to address this issue,

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achieving good performance on six publicly available datasets. Li et al. [22] utilized a 3-D CNN (3D-CNN) to directly process HSIs and extract deep spectral–spatial fusion features, requiring fewer parameters and being lightweight. Yu et al. [23] addressed the issue of overfitting and high training costs caused by the excessive features extracted by 3D-CNNs by designing a novel HSI classification framework using the collaborative effect of 2D-CNN and 3-D convolution layers. First, 2D-CNN extracts spatial features containing abundant spectral information as training channels. Then, 3D-CNN mines band correlation data, simultaneously extracting spatial and spectral features, and providing better fused features for subsequent classification.

Transformers, due to their excellent ability to capture global context, have been widely used in natural language processing and achieved outstanding results. The introduction of Vision Transformer (ViT) has led to breakthrough advancements in image processing, and more researchers are applying the Transformer to HSI classification tasks. Specifically, Hong et al. [15] learned local spectral sequence information from adjacent spectral bands in HSIs to generate grouped spectral embeddings. They also designed a cross-layer skip connection that adaptively learns and fuses cross-layer "soft" residuals, transferring memory-like components from shallow to deep layers. This approach addresses the issue that CNNs cannot effectively mine and represent the sequential properties of spectral features. Sun et al. [24] proposed a spectral–spatial feature tokenized transformer (SSFTT) method to extract spectral–spatial features and high-level semantic features. First, they constructed a spectral–spatial feature extraction module to extract low-level features. Second, a Gaussian-weighted feature tokenizer was introduced to perform feature transformation. Third, the transformed features were input into a Transformer encoder module for feature representation and learning. Finally, a linear layer was used to identify the first learnable token to obtain the sample label. Yang et al. [25] captured subtle spectral differences and conveyed local spatial context information by embedding convolution operations into the Transformer structure, achieving better classification results on four public datasets.

The Mamba model, built upon the SSM architecture, was developed by introducing a selective state-space mechanism. Its outstanding long-sequence modeling capability has garnered significant attention from researchers, leading to a breakthrough in the hyperspectral domain [26]. For instance, He et al. [27] addressed the limited receptive field and quadratic computational complexity of CNN and Transformer models by proposing a novel 3-D spectral–spatial Mamba model. They introduced a 3-D spectral–spatial selective scanning (3DSS) mechanism, which effectively overcame the limitations of the traditional Mamba algorithm, which was restricted to causal sequence modeling and did not adapt well to high-dimensional scenarios. This model enables global spectral–spatial relationship modeling with higher computational efficiency. Pan et al. [28] designed local and global spatial modeling modules (Spa_M) and short-range and long-range spectral dynamic perception modules (Spe_M) based on Mamba. Spa_M captures global spatial semantics while preserving the integrity of local 2-D spatial structures. Spe_M explores the advantages of long-range dependence modeling, achieving more precise short- and long-range spectral feature modeling.

III. Fundamental Concepts and Methods

The proposed HyPyraMamba model, designed for HSI classification, consists of several core modules: patch embedding, PSA, Mamba series modules, AEDC, and the classification head. The specific model structure is shown in Fig. 2.

A. Patch Embedding

Major problems with HSI processing are that the number of input image channels is often very large (e.g., hundreds of spectral bands), making direct convolution or attention operations computationally expensive. To unify the channel dimension and introduce initial feature transformation, the input $X^{(0)} \in \mathbb{R}^{B\times C_{\text{in}}\times H\times W}$ is first mapped to the desired channels using a 1×1 convolution, followed by group normalization on the channels. The specific calculation formula is as follows:

$$Z\left[b,d,i,j\right] = \text{GN}\left(\text{Conv}_{1\times 1}\left(X^{(0)}\right)\right)\left[b,d,i,j\right]. \tag{1}$$

The sigmoid linear unit (SiLU) activation function is then applied

$$\begin{aligned} X^{(1)}\left[b,d,i,j\right] &= \text{SiLU}\left(Z\left[b,d,i,j\right]\right) \\ &= Z\left[b,d,i,j\right]\sigma\left(Z\left[b,d,i,j\right]\right). \end{aligned} \tag{2}$$

Here, b, d, i, and j represent the batch index, hidden channel index, spatial row index, and spatial column index, respectively.

B. PyramidSpectralAttention

HSIs contain both rich spatial texture information and highly correlated spectral information. To simultaneously perform feature extraction in both "multiscale space" and "channel fusion," this article designs the PyramidSpectralAttention module: first, spatial attention is applied at several scales to the input, followed by concatenation and projection of the attention outputs across the channel dimension, thereby achieving the joint fusion of multiscale spatial and spectral information. The specific structure is shown in Fig. 3.

1) Multiscale Downsampling: Let the number of pyramid scales be S, and define the downsampling transformation for each scale index $s = 0, 1, \ldots, S-1$

$$\begin{aligned} &X^{(1,s)}\left[b,d,i_s,j_s\right] \\ &= \begin{cases} X^{(1)}\left[b,d,i_s,j_s\right] & s=0 \\ \dfrac{1}{(2^s)^2}\displaystyle\sum_{u=0}^{2^s-1}\sum_{v=0}^{2^s-1} X^{(1)}\left[b,c,i_s\cdot 2^s+u, j_s\cdot 2^s+v\right] & s\geq 1. \end{cases} \end{aligned} \tag{3}$$

When $s = 0$, no downsampling is applied. When $s \geq 1$, average pooling with a kernel size and stride of 2^s is applied to $X^{(1)}$. Average pooling is chosen because it smooths redundant information across adjacent HSI bands and preserves the continuity of spectral responses. This property aligns with the characteristics of hyperspectral data (i.e., strong spectral correlation and continuous spectra), enabling the downsampled representations to retain informative spectral structures

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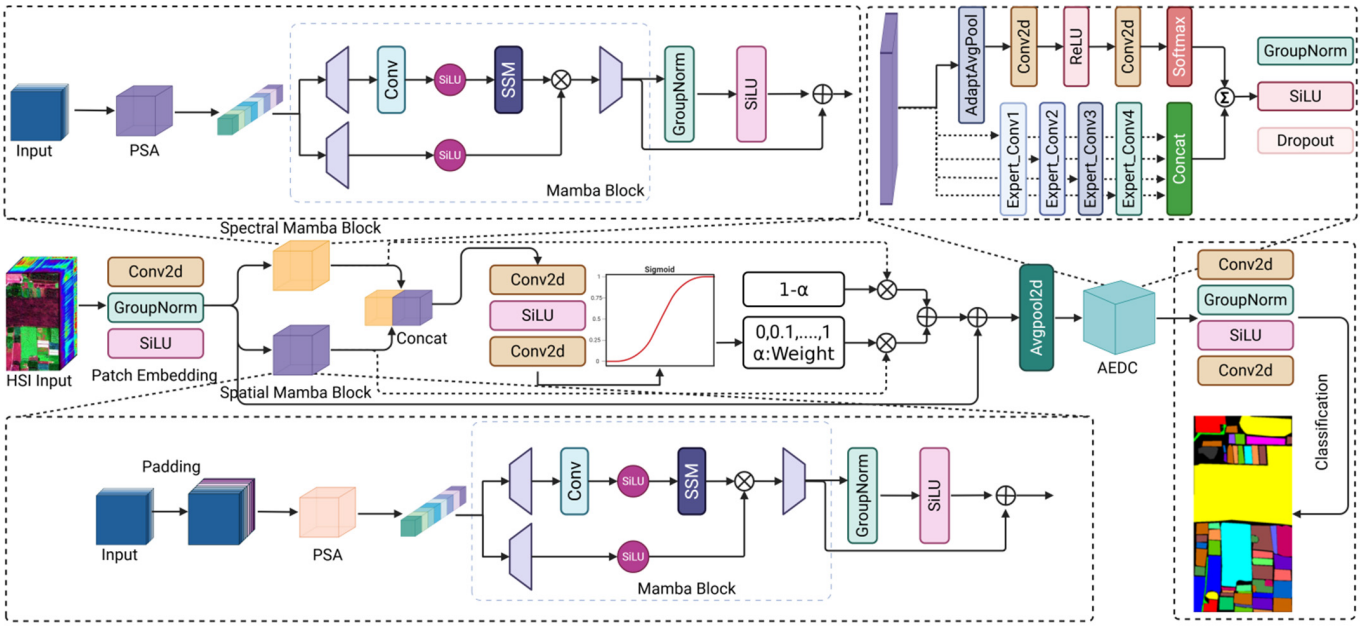


Fig. 2. Overall model architecture.

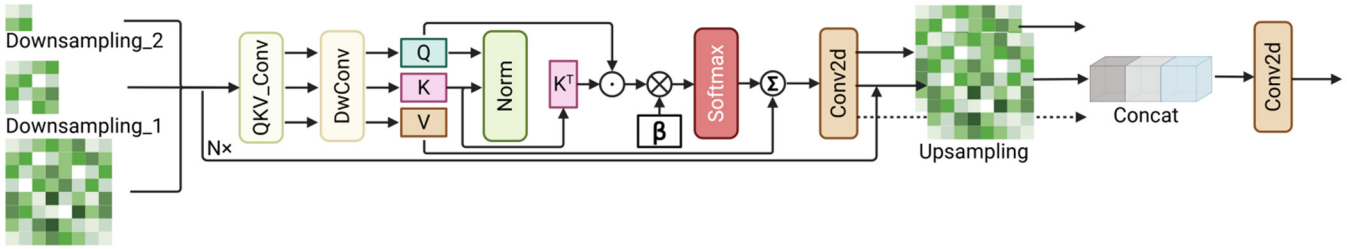


Fig. 3. PSA structure diagram.

while avoiding the spectral distortion often introduced by max pooling. This design supports PSA's objective of capturing multiscale key spectral features sduring downsampling. In addition, the multiscale downsampling is kept moderate in our implementation. The spatial size of the feature maps fed into the PSA module is not smaller than the input patch size (e.g., 9×9), and we only adopt a few small pooling factors (such as 1, 2, and 4) to construct the pyramid. Even at the coarsest scale, the downsampled feature map still contains multiple spatial positions instead of being collapsed into a single pixel, which alleviates the risk of severe information loss on small patches. All scale-specific features are then upsampled back to the original resolution and fused with the original feature via residual connection, so the high-resolution information is preserved, and the multiscale branch mainly serves as a complementary context-enhancement pathway rather than introducing additional noise.

2) Pyramid Attention: For each scale s, multihead spatial attention is applied to the downsampled results. Let X represent the downsampled result for a fixed scale s as an example. First, a 1×1 convolution is used to map the channels from D to $3D$. Then, a depthwise separable 3×3 convolution with a dilation rate of 2 is applied to generate Q, K, and V

$$\widetilde{QKV} = \mathrm{DWConv}\left(\mathrm{Conv}_{1\times1}(X)\right) \in \mathbb{R}^{B\times 3D\times H_s\times W_s}. \quad (4)$$

Divide it evenly into three parts along the channel dimension as follows:

$$Q\left[b,d,i_s,j_s\right] = \widetilde{QKV}\left[b,d,i_s,j_s\right], \quad d=1,\ldots,D \quad (5)$$

$$K\left[b,d,i_s,j_s\right] = \widetilde{QKV}\left[b,D+d,i_s,j_s\right], \quad d=1,\ldots,D \quad (6)$$

$$V\left[b,d,i_s,j_s\right] = \widetilde{QKV}\left[b,2D+d,i_s,j_s\right], \quad d=1,\ldots,D. \quad (7)$$

Then, let the number of heads be H_{head}, with each head having a dimension of $d_{\mathrm{head}} = D/H_{\mathrm{head}}$. Q, K, and V are split into H_{head} subtensors along the channel dimension, and the spatial dimension is flattened to a length of N_s

$$N_s = H_s \times W_s. \quad (8)$$

For the hth head, the query subtensor Q_h is defined as follows:

$$Q_h\left[b,i',p\right] = Q\left[b,(h-1)\,d_{\mathrm{head}}+i',\underbrace{\lfloor (p-1)/W_s \rfloor + 1}_{=i_s},\right.$$

$$\left.\underbrace{(p-1) \bmod W_s + 1}_{=j_s}\right]. \quad (9)$$

Here, p represents the spatial flattening index corresponding to (i_s, j_s), and i' represents the channel subindex within the

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hth head. Similarly, the same splitting and flattening operations are applied to the key and value to obtain

$$K_h\left[b,i',p\right]=K\left[b,(h-1)\,d_{\text{head}}+i',\underbrace{\lfloor(p-1)/W_s\rfloor+1}_{=i_s},\underbrace{(p-1)\bmod W_s+1}_{=j_s}\right] \tag{10}$$

$$V_h\left[b,i',p\right]=V\left[b,(h-1)\,d_{\text{head}}+i',\underbrace{\lfloor(p-1)/W_s\rfloor+1}_{=i_s},\underbrace{(p-1)\bmod W_s+1}_{=j_s}\right]. \tag{11}$$

To improve numerical stability, L2 normalization is applied to Q and K in the last dimension for each head

$$\widetilde{Q_h}\left[b,i',p\right]=\frac{Q_h\left[b,i',p\right]}{\sqrt{\sum_{q=1}^{N_s}Q_h\left[b,i',p\right]^2}} \tag{12}$$

$$\widetilde{K_h}\left[b,i',p\right]=\frac{K_h\left[b,i',p\right]}{\sqrt{\sum_{q=1}^{N_s}K_h\left[b,i',p\right]^2}}. \tag{13}$$

For the hth head, a learnable scaling factor $\tau_h\in\mathbb{R}$ is set. The unnormalized attention score is first calculated as

$$S_h\left[b,p,q\right]=\tau_h\left\langle\tilde{Q}_h\left[b,:,p\right],\tilde{K}_h\left[b,:,q\right]\right\rangle. \tag{14}$$

Here, $\langle\tilde{Q}_h[b,:,p],\tilde{K}_h[b,:,q]\rangle$ represents the dot product of two vectors of length d_{head}, and the index $\tilde{Q}_h[b,:,p]$ refers to all the i' dimension elements corresponding to the pth spatial location for the hth head.

Then, Softmax is applied along the q-dimension to obtain the attention weight matrix A_h, with the specific computation as follows:

$$A_h\left[b,p,q\right]=\frac{\exp\left(S_h\left[b,p,q\right]\right)}{\sum_{q'=1}^{N_s}\exp\left(S_h\left[b,p,q'\right]\right)}. \tag{15}$$

At this point, for a given b, p satisfies the following equation:

$$\sum_{q=1}^{N_s}A_h\left[b,p,q\right]=1. \tag{16}$$

Then, the attention weights A_h are used to perform a weighted sum on the value, obtaining the output for the hth head, with the following computation formula:

$$O_h=\left[b,i',p\right]=\sum_{q=1}^{N_s}A_h\left[b,p,q\right]V_h\left[b,i',q\right]. \tag{17}$$

Next, all H_{head} heads are concatenated along the channel dimension, and the flattened spatial dimension p is reconstructed into a shape of (H_s,W_s). The concatenated tensor is defined as $O^{(s)}$, and its computation is as follows:

$$O^{(s)}\left[b,d,i_s,j_s\right]=O_h\left[b,i',p\right]. \tag{18}$$

Here, $d=(h-1)d_{\text{head}}+i'$ and $p=(i_s-1)W_s+j_s$.

The concatenated tensor is then subjected to a convolutional mapping

$$\tilde{O}^{(s)}\left[b,d,i_s,j_s\right]=\text{Conv}_{1\times1}\left(O^{(s)}\right)\left[b,d,i_s,j_s\right]. \tag{19}$$

3) Multiscale Upsampling and Channel Fusion: For each scale s, when $s=0$, $\hat{O}^{(0)}=\tilde{O}^{(0)}$. When $s\geq1$, upsampling is performed using bilinear interpolation

$$\hat{O}^{(s)}\left[b,d,i,j\right]=\text{Interpolate}\left(\tilde{O}^{(s)},(H,W),\text{bilinear}\right)\left[b,d,i,j\right]. \tag{20}$$

The upsampling results of all scales are concatenated along the channel dimension to obtain O_{cat}

$$O_{\text{cat}}\left[b,:,i,j\right]=\left[\hat{O}^{(0)}\left[b,1:D,i,j\right]\|\hat{O}^{(1)}\left[b,1:D,i,j\right]\|\cdots\|\hat{O}^{(S-1)}\left[b,1:D,i,j\right]\right]. \tag{21}$$

Here, $\|$ indicates the concatenation along the channel direction. Finally, a 1×1 convolution is applied to project the channel count from DS back to D

$$X^{(1a)}\left[b,d,i,j\right]=\text{Conv}_{1\times1}\left(O_{\text{cat}}\right)\left[b,d,i,j\right]. \tag{22}$$

C. Spectra Mamba

First, let the number of tokens be T, and the group size be $G=\lceil D/T\rceil$, $D_{\text{pad}}=TG$. If $D<D_{\text{pad}}$, the input $X^{(1a)}$ is padded along the channel dimension to obtain $X_{\text{pad}}^{(1a)}$. These padded channels do not correspond to real spectral bands and represent neutral responses, while the subsequent group normalization and residual connection allow the network to focus on the original D bands without disrupting their spectral continuity.

In Spectral Mamba, the feature channels are divided into T consecutive groups, each containing G spectral channels. Thus, the effective selective-scanning window corresponds to these G consecutive bands, allowing the model to balance local spectral continuity and long-range spectral dependencies.

Then, $X_{\text{pad}}^{(1a)}$ is flattened by spatial position, and the channel dimension is split. The spatial flattening index p and the corresponding spatial coordinates i and j are computed as follows:

$$p=(i-1)\,W+j \tag{23}$$

$$i=\left\lfloor\frac{p-1}{W}\right\rfloor+1 \tag{24}$$

$$j=(p-1)\bmod W+1. \tag{25}$$

Then, the channels are divided into T segments, each of length G

$$Z\left[b',t,g\right]=X_{\text{pad}}^{(1a)}\left[b,(t-1)\,G+g,i,j\right],\quad\begin{cases}b'=(b-1)\,(HW)+p\\t=1,\ldots,T\\g=1,\ldots,G.\end{cases} \tag{26}$$

Here, b' represents the bth sample's pth spatial pixel in the original batch, t represents the tth token, and g represents the gth dimension within that token.

Next, Z is sent into the Mamba module for sequential modeling, as computed in the following:

$$Z'=\text{Mamba}\left(Z\right). \tag{27}$$

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Then, the spatial and channel layout of Z' is reconstructed

$$Y'\left[b,d,i,j\right]=Z'\left[bHW+p,t,g\right]. \tag{28}$$

Then, group normalization and SiLU activation are applied to Y'

$$\hat{Y}\left[b,d,i,j\right]=\mathrm{SiLU}\left(\mathrm{GN}\left(Y'\left[b,d,i,j\right]\right)\right). \tag{29}$$

Finally, a residual connection is introduced to obtain $\tilde{Y}$, and average pooling is performed to obtain $X^{(2)}$

$$\tilde{Y}\left[b,d,i,j\right]=\begin{cases}\hat{Y}\left[b,d,i,j\right]+X^{(1a)}\left[b,d,i,j\right], & 1\le d\le D\\ \hat{Y}\left[b,d,i,j\right], & D\le d\le D_{\mathrm{pad}}\end{cases} \tag{30}$$

$$X^{(2)}\left[b,d,i',j'\right]=\frac{1}{4}\sum_{u=0}^{1}\sum_{v=0}^{1}\tilde{Y}\left[b,d,2i'-1+u,2j'-1+v\right]. \tag{31}$$

D. Spatial Mamba

In spatial Mamba, each spatial location is modeled as a sequence of length 1, meaning that the effective scanning window corresponds to a single pixel. The global and multiscale spatial context is primarily captured by the preceding PyramidSpectralAttention, while Mamba performs content-adaptive channelwise filtering at each location.

Each channel vector of $X^{(1a)}$ at a spatial position (i,j) is treated as a sequence of length 1, and global spatial sequence modeling is performed to capture spatial context information.

First, each channel is treated as a sequence along with its spatial position. The spatial flattening index is defined as follows:

$$p=(i-1)W+j \tag{32}$$

$$i=\lfloor(p-1)/W\rfloor+1 \tag{33}$$

$$j=(p-1)\bmod W+1. \tag{34}$$

Then, the sequence tensor Z is constructed

$$Z\left[b',1,d\right]=X^{(1a)}\left[b,d,i,j\right] \tag{35}$$

where $b'=(b-1)(HW)+p$. Similarly, Z is sent into the Mamba module for sequential modeling to obtain Z'

$$Z'=\mathrm{Mamba}\left(Z\right). \tag{36}$$

Next, the spatial layout is restored, and Z' is reverted to $\mathbb{R}^{B\times D\times H\times W}$. Let $p=(i-1)W+j$, and thus, the computation of Y' is as follows:

$$Y'\left[b,d,i,j\right]=Z'\left[bHW+p,1,d\right]. \tag{37}$$

Then, group normalization and SiLU activation are applied

$$\hat{Y}\left[b,d,i,j\right]=\mathrm{SiLU}\left(\mathrm{GN}\left(Y'\left[b,d,i,j\right]\right)\right). \tag{38}$$

Finally, a residual connection is introduced to obtain $\tilde{Y}$, and average pooling is performed to obtain the final output $X^{(2)}$

$$\tilde{Y}\left[b,d,i,j\right]=\hat{Y}\left[b,d,i,j\right]+X^{(1a)}\left[b,d,i,j\right] \tag{39}$$

$$X^{(2)}\left[b,d,i',j'\right]=\frac{1}{4}\sum_{u=0}^{1}\sum_{v=0}^{1}\tilde{Y}\left[b,d,2i'-1+u,2j'-1+v\right]. \tag{40}$$

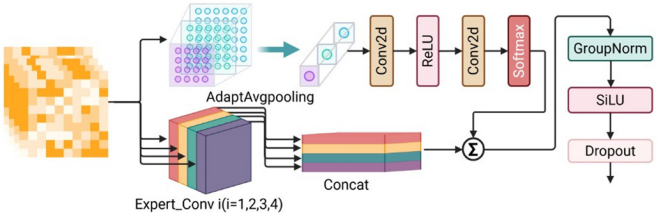


Fig. 4. AEDC structure diagram.

E. Spatial–Spectral Mamba Fusion

First, the spectral and spatial branches are computed separately to obtain the spectral and spatial features

$$\tilde{Y}_{\mathrm{spe}}=\mathrm{SpectralMamba}\left(X^{(1a)}\right) \tag{41}$$

$$\tilde{Y}_{\mathrm{spa}}=\mathrm{SpatialMamba}\left(X^{(1a)}\right). \tag{42}$$

Then, the spectral and spatial features are concatenated along the channel dimension

$$U\left[b,d',i,j\right]=\begin{cases}\tilde{Y}_{\mathrm{spa}}\left[b,d',i,j\right], & d'=1,\ldots,D\\ \tilde{Y}_{\mathrm{spe}}\left[b,d'-D,i,j\right], & d'=D+1,\ldots,2D.\end{cases} \tag{43}$$

Next, two 1×1 convolutions are used to generate a single-channel attention map M. The first 1×1 convolution projects the channel number from $2D$ to D, producing the output

$$V\left[b,d'',i,j\right]=\mathrm{SiLU}\left(\mathrm{Conv}_{1\times1}\left(U\right)\right)\left[b,d'',i,j\right]. \tag{44}$$

Projecting the concatenated features from $2D$ to D ensures dimensional consistency with the outputs of the PSA branches while avoiding redundant duplication of spatial–spectral information. This projection also reduces the computational burden of generating the attention map M and facilitates spatial–spectral fusion.

Then, a second convolution is applied, followed by a Sigmoid activation, to obtain the attention map M:

$$M\left[b,1,i,j\right]=\sigma\left(\mathrm{Conv}_{1\times1}\left(V\right)\right)\left[b,1,i,j\right]. \tag{45}$$

Then, the attention map M is used to perform weighted fusion of the spatial–spectral features

$$Y_{\mathrm{fus}}\left[b,d,i,j\right]=M\left[b,1,i,j\right]\tilde{Y}_{\mathrm{spa}}\left[b,d,i,j\right]+\left(1-M\left[b,1,i,j\right]\right)\tilde{Y}_{\mathrm{spe}}\left[b,d,i,j\right]. \tag{46}$$

Finally, a residual connection is introduced to obtain $\tilde{Y}$, and average pooling is applied

$$\tilde{Y}\left[b,d,i,j\right]=Y_{\mathrm{fus}}\left[b,d,i,j\right]+X^{(1a)}\left[b,d,i,j\right] \tag{47}$$

$$X^{(2)}\left[b,d,i',j'\right]=\frac{1}{4}\sum_{u=0}^{1}\sum_{v=0}^{1}\tilde{Y}\left[b,d,2i'-1+u,2j'-1+v\right]. \tag{48}$$

F. Adaptive Expert Depthwise Conv

We designed the AEDC module to further enhance the model's ability to perceive complex spatial–spectral information and improve classification accuracy, with its specific structure illustrated in Fig. 4. Unlike conventional convolutional networks that struggle to capture such information,

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AEDC links multiple expert kernels with different receptive fields to obtain multiscale spatial–spectral features. The detailed steps of AEDC are as follows.

1) Expert Weights: First, global average pooling is applied to the output of Mamba along the spatial dimension to obtain the channel description z

$$z[b,d,1,1] = \frac{1}{\left(\frac{H}{2}\right)\left(\frac{W}{2}\right)} \sum_{i'=1}^{\frac{H}{2}} \sum_{j'=1}^{\frac{W}{2}} X^{(2)}\left[b,d,i',j'\right]. \tag{49}$$

Next, a 1×1 convolution is applied to map the number of channels from D to D/r, followed by rectified linear unit (ReLU) activation:

$$u\left[b,d',1,1\right] = \mathrm{ReLU}\left(\mathrm{Conv}_{1\times1}(z)\right)\left[b,d',1,1\right]. \tag{50}$$

Then, another convolution is performed to map the number of channels to P, yielding the unnormalized score $\{\alpha_p[b]\}$, which is then passed through Softmax to obtain the expert weights

$$\alpha_p[b] = \mathrm{Conv}_{1\times1}(u)\left[b,p,1,1\right] \tag{51}$$

$$w_p[b] = \frac{\exp\left(\alpha_p[b]\right)}{\sum_{q=1}^{P}\exp\left(\alpha_q[b]\right)}. \tag{52}$$

For each sample b, $\sum_{p=1}^{P} w_p[b] = 1$.

2) Expert Depthwise Separable Convolution: The pth expert is defined as a depthwise separable operator $E_p(\cdot)$, which performs independent convolutions on each input channel

$$E_p\left(X^{(2)}\right)\left[b,d,i',j'\right] = \sum_{(u,v)\in N(i',j')} K_{p,d}(u,v)\, X^{(2)}\left[b,d,i'+u,j'+v\right]. \tag{53}$$

All experts in AEDC adopt depthwise 3×3 kernels (with padding 1). Although the receptive field size is identical, each expert has independent parameters and is adaptively weighted by the expert selector, enabling diverse spatial responses with minimal computational overhead.

3) Weighted Fusion: The outputs of all P experts are weighted and summed according to the weights $w_p[b]$

$$X^{(3)} = \mathrm{Dropout}\left(\mathrm{SiLU}\left(\mathrm{GN}\left(X_{\mathrm{mix}}\right)\right)\right). \tag{54}$$

Then, group normalization, SiLU activation, and dropout are applied sequentially to obtain the final output

$$Z = \mathrm{Conv}_{1\times1}\left(\mathrm{SiLU}\left(\mathrm{GN}\left(\mathrm{Conv}_{1\times1}\left(X^{(3)}\right)\right)\right)\right). \tag{55}$$

Finally, convolution, group normalization, and SiLU are used to produce the classification output.

IV. Experimental Analysis

A. Dataset Description

To evaluate the effectiveness of the proposed method, four representative hyperspectral datasets are employed in this study, namely, IP, WHU-Hi-LongKou (LongKou), Salinas, and Xuzhou HYSPEX (Xuzhou). The IP dataset covers an area of approximately 2.5×2.5 km with a spatial resolution of 20 m, containing about 200 valid bands and 16 land-cover classes. The WHU-Hi-LongKou (LongKou) dataset has a spatial resolution of about 0.463 m, an image size of 550×400, 270 spectral bands, and nine classes with over 120 000 annotated samples. The Salinas dataset provides 200 valid bands with a spatial resolution of about 3.7 m and an image size of 512×217, including 16 crop and land-cover categories. The Xuzhou HYSPEX (Xuzhou) dataset has a spatial resolution of about 0.73 m, an original size of 500×260, and 436 valid bands after preprocessing, with nine classes and approximately 68 877 labeled pixels. These datasets are widely used benchmarks for HSI classification and validation research. The detailed characteristics of the datasets are summarized in Table I.

B. Evaluation Metrics

This study uses three metrics to evaluate classification performance: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient(κ). OA is defined as the ratio of correctly classified pixels to the total number of pixels, used to assess the overall performance of the classification method. AA is the arithmetic mean of classification accuracies across all classes, used to evaluate the algorithm's ability to handle class imbalance. The κ takes into account the effects of random classification, providing an objective evaluation of classification performance. Additionally, extensive experiments are conducted to validate the performance of the proposed model and analyze the impact of different parameter settings on the model's performance.

C. Parameter Configuration

In this study, the model training process is configured as follows: the Adam optimizer is used, with an initial learning rate (lr) of 0.0003 and weight decay set to 1×10^{-5}. The number of training epochs is set to 200, and each experiment is repeated ten times. Random seeds are set from 0 to 9 to ensure the robustness and reproducibility of the experimental results. The cross-entropy loss function is used to compute classification loss, combined with gradient accumulation technology, where gradients are updated every four steps to alleviate GPU memory pressure. Additionally, mixed precision training is enabled during the training process, utilizing PyTorch's automatic mixed precision tools (GradScaler and autocast) to improve training efficiency. Data preprocessing steps include Gaussian filtering for denoising (sigma = 1) and principal component analysis (PCA) for dimensionality reduction, with the number of bands reduced to 30 after PCA. In the experiments, 30 samples per class are used for training, ten samples per class are used for validation, and the remaining samples are used for the test set to evaluate the model's generalization performance. All experiments are conducted under the same hardware configuration to ensure reproducibility, using an NVIDIA A100 PCIe, Intel[1] Xeon[1] Gold 6326 CPU at 2.90 GHz, and 503-GB RAM.

D. Comparative Analysis of the Experimental Results

In this section, we conduct a comprehensive evaluation of the proposed model and compare its performance with several

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TABLE I

DETAILS OF EACH LAND-COVER CLASS OF FOUR DATASETS USED FOR EVALUATION IN THE EXPERIMENTS

Indian Pines (IP)				WHU-Hi-LongKou (LongKou)				Salinas				Xuzhou			
No.	Category	Train	Test	No.	Category	Train	Test	No.	Category	Train	Test	No.	Category	Train	Test
1	Alfalfa	30	16	1	Corn	30	34481	1	Brocoli-green-weeds-1	30	1979	1	Bareland-1	30	26366
2	Corn-notill	30	1398	2	Cotton	30	8344	2	Brocoli-green-weeds-2	30	3696	2	Bareland-2	30	3997
3	Corn-mintill	30	800	3	Sesame	30	3001	3	Fallow	30	1946	3	Crops-1	30	2753
4	Corn	30	203	4	Broad-leaf soybean	30	63182	4	Fallow-rough-plow	30	1364	4	Crops-2	30	5184
5	Grass-pasture	30	453	5	Narrow-leaf soybean	30	4121	5	Fallow-smooth	30	2648	5	Coals	30	13154
6	Grass-tree	30	700	6	Rice	30	11824	6	Stubble	30	3929	6	Concrete	30	2406
7	Grass-pasture-mowed	30	10	7	Water	30	67026	7	Celery	30	3549	7	Trees	30	6960
8	Hay-windrowed	30	448	8	Roads and houses	30	7094	8	Grapes-untrained	30	11241	8	Lake	30	4747
9	Oats	30	10	9	Mixed weed	30	5199	9	Soil-vineyard-develop	30	6173	9	House-Roof	30	3040
10	Soybean-notill	30	942					10	Corn-senesced-green-weeds	30	3248				
11	Soybean-minitill	30	2425					11	Lettuce-romaine-4wk	30	1038				
12	Soybean-clean	30	563					12	Lettuce-romaine-5wk	30	1897				
13	Wheat	30	175					13	Lettuce-romaine-6wk	30	886				
14	Woods	30	1235					14	Lettuce-romaine-7wk	30	1040				
15	Buildings-Grass-Trees-Drives	30	356					15	Vinyard-untrained	30	7238				
16	Stone-Steel-Towers	30	63					16	Vinyard vertical trellis	30	1770				
Total	-	480	9800	-	-	480	204272	-	-	480	53642	-	-	480	68607

TABLE II

COMPARATIVE EXPERIMENTAL RESULTS ON THE IP DATASET. (PREFERABLY BOLD, SECOND UNDERSCORE. THESE ANNOTATIONS APPLY TO ALL TABLES)

No.	Based CNN			Based Transformer			Based Mamba		
	A2S2KResNet 2020	CSCANet 2025	SACNet 2021	DiffFormer 2025	SVAFormer 2024	SQSFormer 2024	MambaHSI 2025	SSMamba 2024	HyPyraMamba Ours
1	45.97±27.80	**100.00±0.00**	**100.00±0.00**	83.33±2.75	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
2	62.97±9.39	47.79±4.38	59.47±9.44	46.61±3.58	75.53±3.65	**86.05±3.26**	72.31±6.81	79.68±3.70	84.88±4.59
3	53.02±12.35	55.19±2.25	70.33±9.09	38.10±3.02	75.75±4.25	91.87±0.25	79.75±7.92	88.75±2.62	**91.93±3.61**
4	63.35±11.41	64.73±3.83	97.87±3.37	58.38±2.73	96.13±0.36	**100.00±0.00**	99.59±1.06	99.22±0.78	98.73±1.66
5	91.68±5.51	90.64±0.98	86.93±6.67	83.07±2.12	93.59±0.05	94.26±1.32	85.58±5.79	92.89±3.84	**94.93±2.91**
6	92.31±5.00	98.23±0.64	96.77±1.20	97.68±3.23	94.00±1.26	98.42±0.58	96.75±2.99	**99.54±0.35**	99.20±0.55
7	0.00±0.00	0.00±0.00	80.00±30.55	0.00±0.00	**100.00±0.00**	**100.00±0.00**	96.67±10.00	**100.00±0.00**	**100.00±0.00**
8	95.75±6.81	93.19±2.20	99.73±0.37	91.55±4.46	99.55±0.02	**100.00±0.00**	99.79±0.33	**100.00±0.00**	**100.00±0.00**
9	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	59.86±17.94	62.71±3.24	68.42±13.8	54.94±1.92	85.24±6.32	90.33±2.23	78.70±7.01	86.94±5.91	**90.97±4.16**
11	78.01±6.90	45.52±2.59	53.81±16.56	54.99±3.96	67.75±9.20	78.68±6.98	67.88±7.70	83.56±5.53	**90.18±3.36**
12	51.14±14.32	44.62±4.66	55.44±11.03	40.69±2.65	80.46±6.20	**96.80±0.36**	82.98±7.14	89.09±3.79	89.80±4.27
13	80.49±18.15	99.03±0.37	98.67±1.01	96.36±2.84	99.42±0.03	99.42±0.02	99.76±0.40	**99.88±0.22**	99.70±0.49
14	90.48±5.26	85.28±1.76	84.98±7.42	73.88±4.63	91.49±0.65	98.62±0.45	94.49±2.85	98.25±1.24	**98.71±1.60**
15	70.76±16.80	63.37±5.10	85.06±4.83	48.55±0.36	93.25±0.23	**99.71±0.06**	96.33±3.81	99.21±0.69	98.27±3.33
16	44.96±14.83	**100.00±0.00**	99.81±0.57	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.68±3.96	99.68±0.63	**100.00±0.00**
OA(%)	68.53 ± 6.84	63.76 ± 1.13	71.20±4.74	60.79±5.13	81.86±4.26	90.39±5.36	81.52±2.85	89.60±1.96	**92.40±1.15**
AA(%)	61.30 ± 4.16	65.64 ± 1.01	82.49±2.62	60.51±3.23	90.76±0.05	94.88±1.25	89.95±1.51	94.79±0.76	**95.44±0.70**
κ×100	64.76 ± 7.18	59.21 ± 1.24	82.15±4.94	55.57±12.28	79.47±10.36	89.08±4.25	90.15±2.41	88.16±2.21	**95.80±1.81**

recent state-of-the-art methods. The competing approaches include advanced CNN-based models (A2S2KResNet [29], CSCANet [30], and SACNet [31]), optimized Transformer-based HSI classification models (DiffFormer [32], SVAFormer [33], and SQSFormer [34]), and the latest Mamba-based models for HSI classification (MambaHSI [35] and SSMamba [36]). Experiments are carried out on four representative datasets, namely, IP, LongKou, Salinas, and Xuzhou, with results summarized in Tables II–V. The overall comparison demonstrates that the proposed model (Ours) consistently achieves superior performance across all datasets, significantly outperforming existing CNN-, Transformer-, and Mamba-based methods, thereby validating its effectiveness and robustness for HSI classification.

Specifically, on the IP dataset, our model excels in recognizing complex categories. For instance, the accuracy for Class 7 reaches 100.00%, significantly outperforming other comparison models. Furthermore, in Classes 3 and 6, our model achieves high accuracies of 91.73% and 99.20%, respectively, significantly outperforming the CNN-based A2S2KResNet and Transformer-based DiffFormer models. This advantage mainly stems from the innovative combination of the PSA mechanism and the spatial–spectral Mamba module, which effectively enhances the perception and discrimination ability for complex spatial–spectral structures. For the LongKou dataset, the spectral features between categories are highly similar, and some categories. In contrast, our model achieves high accuracies of 97.99%, 97.63%, and 98.99% for cat-

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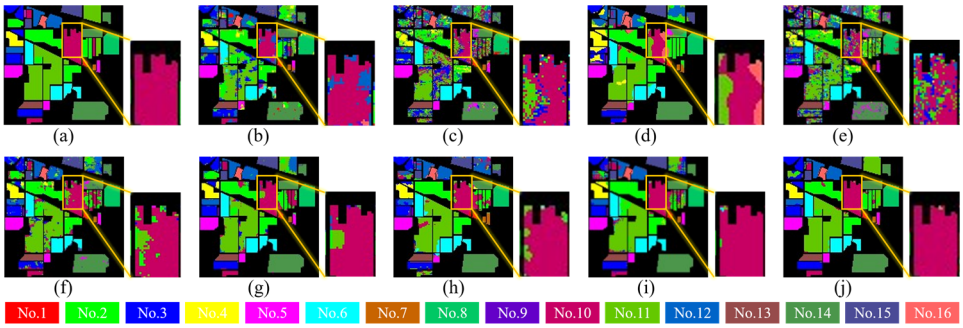


Fig. 5. Visual comparison results on the IP dataset. (a) Reference map. (b) A2S2KResNet. (c) CSCANet. (d) SACNet. (e) DiffFormer. (f) SVAFormer. (g) SQSFormer. (h) MambaHSI. (i) SSMamba. (j) Ours.

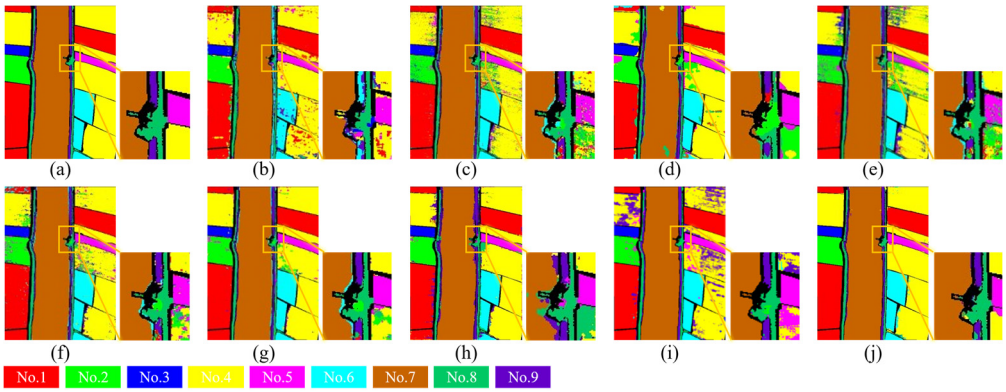


Fig. 6. Visual comparison results on the LongKou dataset. (a) Reference map. (b) A2S2KResNet. (c) CSCANet. (d) SACNet. (e) DiffFormer. (f) SVAFormer. (g) SQSFormer. (h) MambaHSI. (i) SSMamba. (j) Ours.

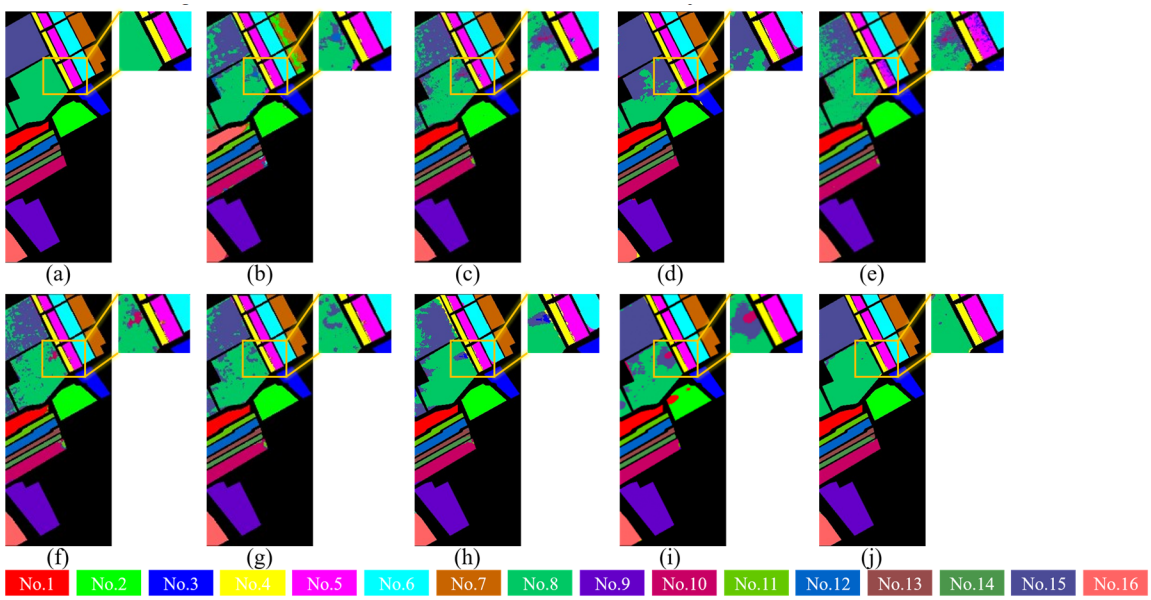


Fig. 7. Visual comparison results on the Salinas dataset. (a) Reference map. (b) A2S2KResNet. (c) CSCANet. (d) SACNet. (e) DiffFormer. (f) SVAFormer. (g) SQSFormer. (h) MambaHSI. (i) SSMamba. (j) Ours.

egories 2, 5, and 7, respectively, with an OA of 97.59%, clearly outperforming existing models. This indicates that the adaptive expert depthwise separable convolution structure of our model better captures and differentiates subtle spectral differences between similar land cover types. On the Salinas dataset, the spectral features of the crop categories are clear, but subtle differences are significant. Particularly in the challenging-to-distinguish Category 8, our model outperforms other methods with an accuracy of 93.57%. Additionally, the model performs outstandingly on OA = 97.39% and AA

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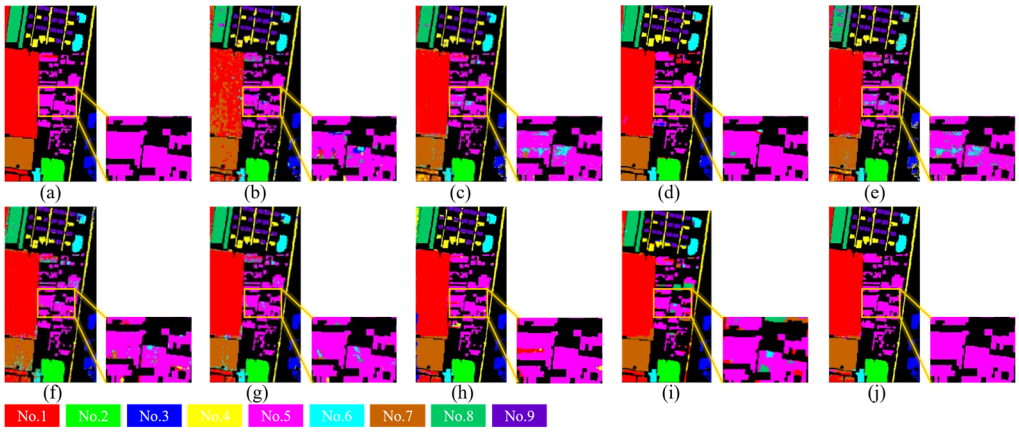


Fig. 8. Visual comparison results on the Xuzhou dataset. (a) Reference map. (b) A2S2KResNet. (c) CSCANet. (d) SACNet. (e) DiffFormer. (f) SVAFormer. (g) SQSFormer. (h) MambaHSI. (i) SSMamba. (j) Ours.

TABLE III
COMPARATIVE EXPERIMENTAL RESULTS ON THE LONGKOU DATASET

No.	Based CNN			Based Transformer			Based Mamba		
	A2S2KResNet 2020	CSCANet 2025	SACNet 2021	DiffFormer 2025	SVAFormer 2024	SQSFormer 2024	MambaHSI 2025	SSMamba 2024	HyPyraMamba Ours
1	79.76±29.56	98.93±0.31	92.97±5.53	98.67±2.75	97.29±0.36	**99.48±0.02**	96.35±1.51	95.81±5.94	98.79±1.17
2	69.98±30.76	91.36±2.34	90.59±7.02	80.51±3.58	90.34±0.45	93.81±0.25	97.09±1.55	78.75±30.91	**97.99±2.05**
3	73.56±26.67	99.39±0.50	97.18±1.83	99.03±3.02	99.13±0.31	98.00±0.15	**99.71±0.35**	98.26±2.16	99.66±0.56
4	96.33±2.90	86.45±2.28	91.15±2.83	79.53±2.73	88.93±4.23	95.65±1.01	94.57±2.84	76.97±24.70	**95.72±1.64**
5	66.43±23.15	95.56±0.60	95.76±3.65	87.47±2.12	94.95±0.58	**98.54±0.36**	96.67±1.88	80.62±31.53	97.63±1.52
6	73.71±20.75	**99.31±0.20**	93.83±3.80	99.09±3.23	98.97±0.12	99.20±0.02	96.25±1.77	98.14±2.94	98.57±1.24
7	88.77±29.65	98.94±0.21	94.04±5.33	98.50±2.19	98.75±0.19	97.77±0.36	94.12±2.03	99.96±0.03	**98.99±1.01**
8	68.39±26.79	91.31±2.26	83.87±7.88	91.77±4.46	94.43±0.66	94.44±1.36	87.08±4.81	92.70±3.37	**94.58±1.62**
9	53.05±16.11	95.41±0.95	93.11±4.35	86.82±4.82	85.24±4.25	94.65±0.98	90.36±3.81	**96.56±1.84**	94.22±2.13
OA(%)	80.93 ± 15.33	94.37 ± 0.89	92.52±2.32	91.22±2.14	94.57±3.33	92.72±3.36	94.67±1.35	90.42±9.31	**97.59±0.48**
AA(%)	74.44 ± 11.57	95.18 ± 0.77	92.50±1.45	91.27±0.85	94.22±1.23	94.49±4.12	94.69±1.00	90.86±6.12	**97.35±0.42**
κ×100	76.80 ± 17.27	92.72 ± 1.14	92.75±2.78	88.72±1.28	92.96±1.02	90.64±1.26	95.00±2.17	88.09±11.14	**97.12±1.12**

= 98.51%, fully demonstrating the model's high sensitivity and robustness in recognizing subtle spatial–spectral features. The Xuzhou dataset mainly consists of complex urban scenes with high background noise and mixed land cover types, making classification difficult. Our model achieves the highest accuracies of 99.80%, 97.76%, 97.61%, and 99.80% in the difficult categories 3–6, with an OA of 96.05%. These results clearly demonstrate the model's strong generalization ability and robustness in high-noise and complex scenarios, significantly outperforming traditional CNN and existing Transformer and Mamba structures. Therefore, the proposed model, with its innovative design of the PSA mechanism, the spatial–spectral dual-path Mamba module, and the adaptive expert convolution structure, exhibits significant advantages and stability across hyperspectral datasets with varying types and complexities, validating the effectiveness of the model architecture design and its strong adaptability to complex data scenarios.

E. Visualization Analysis

To visually demonstrate the superiority of the proposed algorithm, Figs. 5–8 display the comparative experimental results and zoomed-in view images for the IP, LongKou, Salinas, and Xuzhou datasets, respectively. From the visualized images, it is clear that Ours outperforms other comparison methods on all four datasets, showing clearer class boundaries and less classification noise. Specifically, on the IP dataset, our model significantly reduces confusion and misclassification between land cover classes, especially achieving more accurate classification in small area categories and boundary regions, reflecting the model's efficiency in extracting local region information. The visualization results for the LongKou dataset show that our model effectively avoids the mixed classification problem at category boundaries, making the boundaries between different categories clearer. The adaptive expert depthwise separable convolution structure of the model better captures and differentiates subtle spectral differences between similar land cover types. This structure improves the model's sensitivity to subtle differences in land cover categories and enhances overall classification stability and accuracy. This advantage is crucial for improving land cover classification accuracy in practical hyperspectral data applications. On the Salinas dataset, the classification within crop planting areas is clear and uniform. Our model effectively reduces local confusion and fragmented classification results, demonstrating its advantage in maintaining regional consistency. The Xuzhou dataset contains a complex urban environmental background,

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TABLE IV
COMPARATIVE EXPERIMENTAL RESULTS ON THE SALINAS DATASET

No.	Based CNN			Based Transformer			Based Mamba		
	A2S2KResNet 2020	CSCANet 2025	SACNet 2021	DiffFormer 2025	SVAFormer 2024	SQSFormer 2024	MambaHSI 2025	SSMamba 2024	HyPyraMamba Ours
1	87.62±29.46	99.99±0.02	97.95±3.49	99.04±0.75	98.89±0.08	**100.00±0.00**	99.27±0.95	**100.00±0.00**	**100.00±0.00**
2	96.71±7.94	99.35±0.23	99.67±0.50	99.97±0.58	99.97±0.01	**100.00±0.00**	99.82±0.31	98.14±3.71	**100.00±0.00**
3	94.74±4.48	99.95±0.05	93.97±9.20	99.02±0.02	**100.00±0.00**	99.95±0.12	**100.00±0.00**	99.98±0.04	**100.00±0.00**
4	90.68±5.09	99.76±0.16	96.26±5.49	99.93±0.73	99.71±0.12	**100.00±0.00**	99.60±0.45	99.87±0.14	**100.00±0.00**
5	92.95±8.24	98.46±0.19	95.59±4.29	92.65±2.12	98.04±0.35	97.66±0.35	95.70±1.38	**98.61±1.07**	98.24±0.77
6	97.88±1.97	99.99±0.01	97.98±1.66	**100.00±0.00**	99.75±0.04	**100.00±0.00**	98.97±0.86	**100.00±0.00**	**100.00±0.00**
7	97.58±2.07	99.75±0.19	97.74±1.56	**100.00±0.00**	99.97±0.02	**100.00±0.00**	99.88±0.23	99.99±0.01	**100.00±0.00**
8	89.26±5.95	81.45±1.32	68.05±18.19	83.61±4.46	83.04±0.48	82.95±2.26	85.72±5.35	78.76±17.21	**93.57±4.20**
9	93.03±14.74	99.99±0.00	99.42±1.51	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.99±0.01	**100.00±0.00**	**100.00±0.00**
10	96.25±2.71	96.28±0.32	93.05±8.41	96.66±1.92	94.33±1.28	95.84±0.36	95.81±1.67	**98.77±0.61**	98.13±1.43
11	94.03±4.71	99.18±0.19	99.89±0.24	97.28±1.96	99.71±0.06	99.81±0.02	99.85±0.25	99.90±0.15	**99.96±0.09**
12	94.86±6.07	99.92±0.12	97.10±3.63	99.79±0.65	**100.00±0.00**	**100.00±0.00**	98.47±1.06	**100.00±0.00**	**100.00±0.00**
13	95.89±4.18	99.06±1.21	97.39±3.05	99.77±0.84	**100.00±0.00**	**100.00±0.00**	99.28±0.95	99.95±0.09	97.45±6.85
14	94.34±10.45	95.96±0.89	95.90±5.14	99.13±0.63	99.23±0.55	99.90±0.04	96.25±3.57	**100.00±0.00**	97.76±1.70
15	73.14±16.51	79.81±1.22	77.48±22.73	66.41±0.36	80.35±2.83	89.26±1.25	85.17±8.95	85.54±20.74	**93.48±6.82**
16	91.24±14.64	99.06±0.10	96.46±3.69	98.81±0.44	99.38±0.42	99.55±0.12	99.07±0.71	**99.81±0.20**	99.16±0.40
OA(%)	89.72 ± 5.03	92.87 ± 0.28	88.49±3.11	90.26±1.26	93.24±0.69	94.59±0.56	94.23±1.16	93.31±2.99	**97.39±1.53**
AA(%)	92.51 ± 3.57	96.75 ± 0.17	93.99±1.33	91.26±0.54	97.02±0.13	97.81±0.24	97.05±0.60	97.46±1.04	**98.51±0.67**
κ×100	88.61 ± 5.50	92.07 ± 0.31	94.00±2.29	95.75±0.68	92.48±0.28	93.98±0.36	96.33±2.09	92.57±3.31	**98.00±1.15**

TABLE V
COMPARATIVE EXPERIMENTAL RESULTS ON THE XUZHOU DATASET

No.	Based CNN			Based Transformer			Based Mamba		
	A2S2KResNet 2020	CSCANet 2025	SACNet 2021	DiffFormer 2025	SVAFormer 2024	SQSFormer 2024	MambaHSI 2025	SSMamba 2024	HyPyraMamba Ours
1	98.12±1.20	88.29±1.21	86.74±2.33	89.04±2.75	91.90±0.38	92.63±1.33	89.30±1.90	87.22±11.11	**92.71±2.00**
2	96.87±4.80	97.33±0.45	96.59±2.38	96.26±1.58	96.09±1.23	97.44±1.25	98.85±0.48	**99.25±0.37**	99.11±0.74
3	76.66±17.31	83.94±2.30	94.76±2.91	73.28±3.02	97.31±1.02	97.96±0.25	99.44±0.64	95.92±4.35	**99.80±0.30**
4	87.15±7.02	92.53±0.86	88.40±5.94	91.86±2.73	89.31±0.25	96.46±0.33	93.31±1.45	93.85±5.85	**97.76±1.53**
5	94.23±3.22	86.36±5.60	92.34±6.40	90.48±2.12	88.30±0.33	93.44±0.28	95.87±2.83	90.83±7.97	**97.61±1.91**
6	71.90±22.72	92.31±1.96	95.12±3.08	78.80±6.23	81.50±1.36	93.10±0.10	99.48±0.30	91.79±6.81	**99.80±0.35**
7	77.53±11.92	90.38±2.27	98.69±1.55	93.24±2.19	88.52±1.28	94.36±3.01	97.54±1.93	**99.13±0.81**	98.83±2.04
8	81.95±14.43	95.19±1.30	96.83±2.51	80.83±4.46	97.93±0.16	99.34±0.03	99.76±0.30	97.77±3.58	**99.84±0.28**
9	80.49±31.64	99.63±0.16	89.90±3.45	**99.87±0.02**	97.56±0.35	95.85±1.22	92.87±1.02	97.75±1.17	95.41±1.82
OA(%)	88.26 ± 3.78	89.92 ± 1.22	91.18±1.63	89.30±1.28	91.43±0.48	94.37±0.88	93.90±0.98	92.03±3.87	**96.05±0.79**
AA(%)	84.99 ± 6.01	91.77 ± 0.96	93.26±1.04	88.18±0.68	92.05±6.22	95.62±2.26	96.27±0.49	94.83±0.72	**97.84±0.43**
κ×100	85.41 ± 4.63	87.43 ± 1.50	93.75±3.11	86.58±2.03	89.24±5.36	92.92±2.00	93.00±2.38	90.11±4.60	**97.25±1.56**

TABLE VI
ABLATION EXPERIMENTAL ANALYSIS

Spe_M	Spa_M	AEDC	PSA	Fusion	IP	LongKou	Salinas	XuZhou
×	✓	✓	✓	×	89.73±2.40	96.89±0.99	95.88±0.67	95.09±1.11
✓	×	✓	✓	×	91.84±1.40	97.29±0.91	96.43±0.83	95.51±0.74
✓	✓	×	✓	✓	73.30±2.63	91.35±1.88	91.18±0.69	92.08±0.85
✓	✓	✓	×	✓	83.95±1.90	93.17±3.26	93.30±1.05	93.74±2.17
×	✓	✓	×	×	83.80±1.89	93.05±3.60	93.43±0.86	93.93±1.92
✓	×	✓	×	×	83.73±1.63	94.03±2.84	93.42±0.75	93.35±2.18
✓	✓	✓	✓	✓	92.40±1.15	97.59±0.48	97.39±1.53	96.05±0.79

with significant overlaps between categories. The introduction of the spatial–spectral dual-path Mamba module effectively addresses this issue, maintaining category consistency and stability even in complex backgrounds, and effectively avoiding significant misclassifications and regional confusion. These visualization results demonstrate the generality and robustness of the proposed model in different application scenarios, confirming its practical value in HSI classification tasks.

F. Ablation Experimental Analysis

To further validate the effectiveness of each key module in the proposed model, comprehensive ablation experiments were conducted, and the results are shown in Table VI. In this experiment, we sequentially verified the contributions of the Spe_M, Spa_M, AEDC, PSA, and the spatial–spectral fusion module to classification performance. The results in the table show that when only the Spe_M or Spa_M

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TABLE VII
COMPUTATIONAL COMPLEXITY COMPARISON ANALYSIS

Metrics	Based CNN			Based Transformer			Based Mamba		
	A2S2KResNet	CSCANet	SACNet	DiffFormer	SVAFormer	SQSFormer	MambaHSI	SSMamba	Ours
Parameters(M)	0.37	7.37	0.21	0.11	0.12	0.17	0.44	0.54	1.21
FLOPs(G)	0.16	0.69	3.3	0.84	38.93	6.68	107.5	72.11	94.26

modules were removed, the model's classification accuracy decreased, with a more significant drop when Spe_M was removed, indicating that the spectral Mamba module is crucial for fully exploiting spectral feature information and improving classification accuracy. The Spa_M module, on the other hand, effectively enhances the utilization efficiency of spatial structural information, and its removal also led to a noticeable performance drop. This suggests that the collaboration between these two modules is critical for the model's overall performance. When the AEDC module was removed, the classification accuracy of the datasets dropped significantly, especially on the IP dataset (decreasing from 92.40% to 73.30%), indicating that the introduction of this module has a clear advantage in handling complex spatial–spectral features and enhancing the ability to express local features. After removing the PSA module, the overall classification performance decreased significantly, especially on the LongKou and IP datasets. This result shows that the PSA module plays a crucial role in multiscale information extraction and improving category discrimination. When the fusion module was absent, classification accuracy across datasets was generally lower, highlighting the important role of the spatial–spectral fusion strategy in enhancing feature expression and improving the fine-grained classification ability. The complete model, which integrates all modules, achieved the highest classification accuracy across all four datasets, with the most significant improvement observed on the IP dataset. This confirms that the proposed modules have a good synergistic effect and can effectively enhance the overall performance and robustness of the model. Therefore, the ablation experimental results validate the necessity and effectiveness of each module proposed in this study, demonstrating the rationality and effectiveness of the model's structural design.

G. Computational Complexity Analysis

To comprehensively evaluate the efficiency of the proposed model, we compare it with several existing methods based on different architectures (CNN, Transformer, and Mamba) in terms of two key metrics: the number of parameters and floating point operations per second (FLOPs). Table VII summarizes the results. HyPyraMamba contains 1.21M parameters, which is a larger number than those of lightweight models such as DiffFormer (0.11M) and A2S2KResNet (0.37M). However, the remaining is a relatively small magnitude which is acceptable for practical HSI classification. Regarding computational complexity, our model requires 94.26G FLOPs, which is higher than the CNN- and Transformer-based baselines, reflecting the additional cost introduced by the PSA, AEDC, and dual Mamba branches. In return, HyPyraMamba achieves the best OA/AA among all compared methods on the four datasets (see Tables II–V), especially on challenging and spectrally similar classes. Compared with other strong Mamba-based models (MambaHSI: 107.5G ad SSMamba: 72.11G), HyPyraMamba operates at a comparable complexity level while providing higher classification accuracy. Therefore, our model should be regarded as a high-performance architecture that offers a reasonable accuracy–efficiency tradeoff, rather than an extremely lightweight solution.

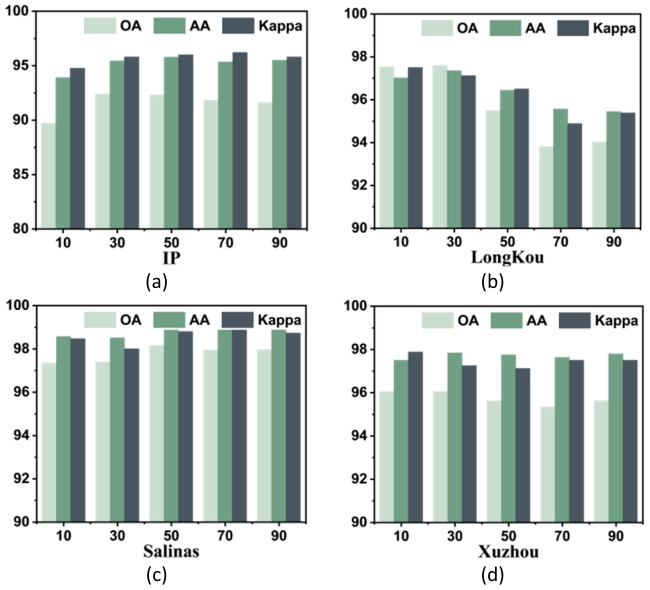


Fig. 9. Results of different PCA analyses. (a) IP dataset, (b) longKou, (c) salinas, (d) xuzhou.

H. Training Parameter Analysis

1) PCA Analysis: As shown in Fig. 9(a)–(d), this study conducted a detailed comparative experiment on the impact of different PCA-reduced feature dimensions on the model's classification accuracy to verify the robustness and applicability of the proposed model at different spectral dimensions. According to the experimental results on the IP dataset, as the PCA dimension increases from 10 to 50, both the OA and AA show a significant upward trend. When the dimension reaches 30, OA is maximized, and κ also shows high performance. However, when the dimension is further increased to 70 and 90, both OA and κ decrease, indicating that appropriate dimensionality selection is crucial for removing spectral redundancy and retaining meaningful information in the IP dataset. The LongKou dataset shows that when the PCA dimension is low (10)–(30), the model performs best in classification, with a dimension of 30 achieving the highest OA and a relatively high AA. This reflects the model's ability to effectively capture key

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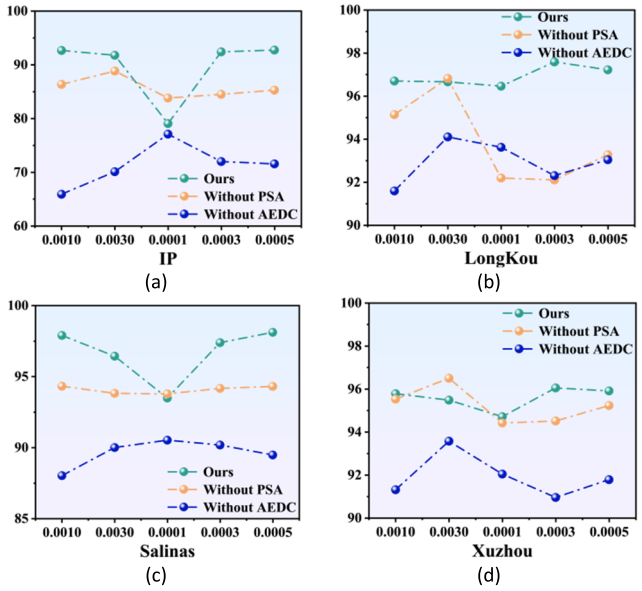


Fig. 10. Results of different lr analyses. (a) IP dataset, (b) longKou, (c) salinas, (d) xuzhou.

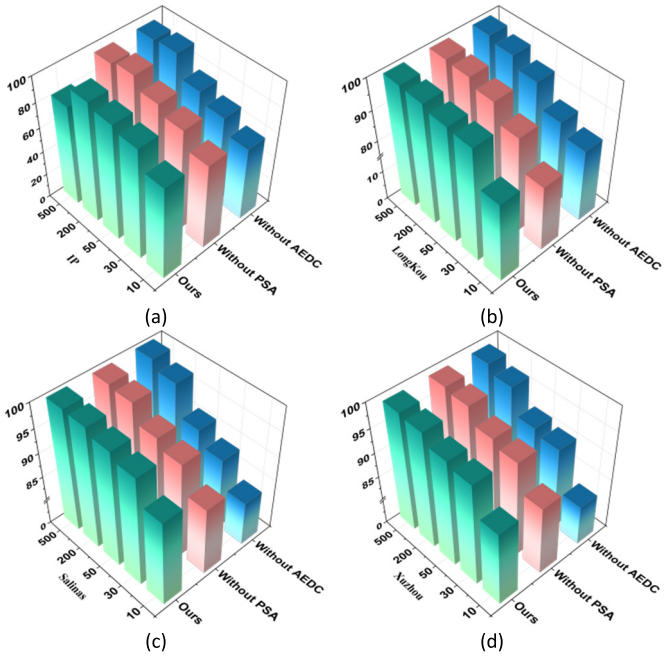


Fig. 11. Results of different training sample analyses. (a) IP dataset, (b) longKou, (c) salinas, (d) xuzhou.

features with fewer dimensions. As the dimension increases further to 70 and 90, the model's performance decreases, suggesting that the LongKou dataset contains some spectral redundancy, and excessively high dimensions may introduce noise or redundant features. On the Salinas dataset, the model performs best with a PCA dimension of 50 in terms of overall classification performance. When the dimension is too low or too high, classification accuracy slightly decreases but remains at a high level overall. This suggests that the feature bands in the Salinas dataset are more uniformly correlated, and a moderate feature dimension is more beneficial for the model in extracting accurate spatial–spectral information. The experimental results on the Xuzhou dataset show that when the dimensions are 10 and 30, the model achieves the highest OA and AA, indicating that the model effectively captures the main spectral features in complex urban scenes at lower dimensions. When the dimension increases beyond 50, the model's accuracy slightly decreases, reflecting redundancy in the spectral feature space of the urban scene, further supporting the model's robustness under appropriate dimensionality selection. Finally, this study selects a PCA dimension of 30 as the optimal feature dimension. This shows that this dimensionality setting effectively reduces the data dimensions, removes redundant spectral information, and maximally retains the key features in hyperspectral images, improving the model's generalization performance and stability.

2) Learning Rate Analysis: To further optimize model performance, this study systematically examined the impact of varying lr settings on the classification performance of three distinct models (Ours, Without PSA, and Without AEDC) across the IP, LongKou, Salinas, and Xuzhou datasets, using OA as an example metric. The experimental results are presented in Fig. 10(a)–(d). Results from the IP dataset reveal that overall error (OA) decreases when lr is set excessively high or low. However, when lr values range between 0.0003 and 0.0005, OA remains consistently high with minimal fluctuation, demonstrating excellent stability. On the LongKou dataset, the overall performance of Ours across different lr values tends to be consistent. Nevertheless, it can still be observed that OA reaches its peak at 0.0003, significantly outperforming Without PSA and Without AEDC. This indicates that this lr value possesses both good generalization capability and training stability on the LongKou dataset. For the Salinas dataset, all three models achieved relatively good results. The Xuzhou dataset maintained high accuracy with minimal fluctuation, demonstrating strong adaptability in complex urban environments. Consequently, this study ultimately selected 0.0003 as the lr value for the HyPyraMamba model. This setting achieved high classification accuracy, low variance, and favorable convergence speed across all four datasets, fully validating the rationality and robustness of this parameter choice.

3) Training Sample Analysis: To further investigate the impact of training sample size on the classification performance of the HyPyraMamba model, comparative experiments were conducted across four datasets with varying numbers of training samples. The experimental results are presented in Fig. 11(a)–(d).

The experimental results across datasets reveal that when the number of training samples is low (e.g., ten samples), the OA of all three models is relatively low and exhibits significant fluctuations. This indicates that the scarcity of hyperspectral data samples significantly impacts the models' learning capacity and generalization performance. When the sample size increases to 30, the OA value improves markedly, demonstrating the models' strong adaptability to limited training data. Further increases to 50 and 200 samples yield performance improvements although the rate of gain slows. At 500 samples, while OA remains high, certain datasets like IP show noticeable decline, suggesting excessive samples may induce overfitting or introduce redundant noise. Results across all four datasets collectively indicate that 30

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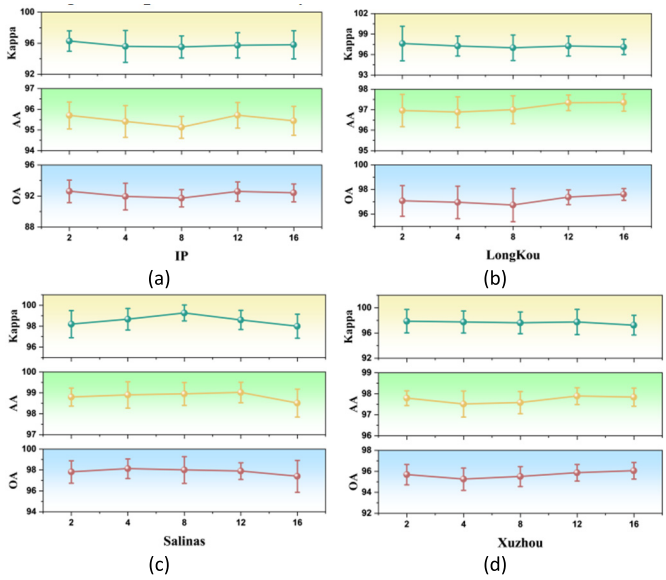


Fig. 12. Results of different state analyses. (a) IP dataset, (b) longKou, (c) salinas, (d) xuzhou.

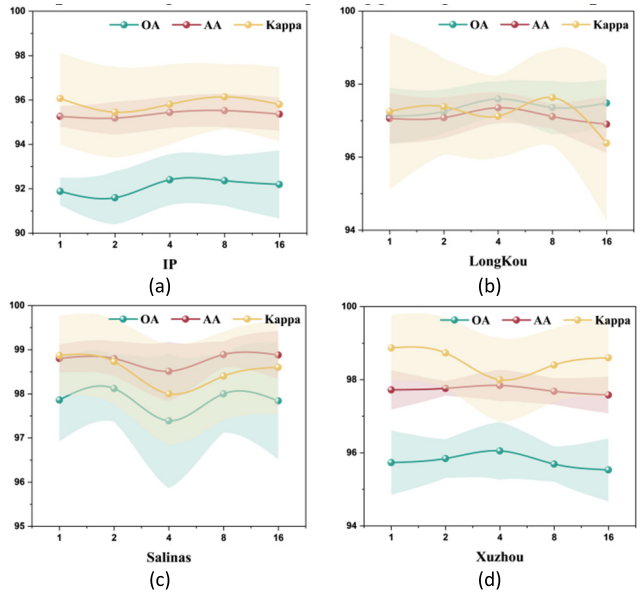


Fig. 13. Results of different head analyses. (a) IP dataset, (b) longKou, (c) salinas, (d) xuzhou.

training samples strike a favorable balance between achieving high classification accuracy, controlling annotation costs, and maintaining model performance. Compared to fewer samples, 30 samples markedly reduce noise interference and variability; compared to larger samples, they avoid unnecessary computational costs and potential overfitting risks. Consequently, this study definitively selects 30 training samples as the optimal model training configuration, uniformly applied across all subsequent experiments. This choice not only demonstrates the robustness of the HyPyraMamba model in small-sample scenarios but also highlights its broad application potential in hyperspectral classification tasks.

4) State Analysis: To further explore the impact of the state dimension parameter in the Mamba module on model performance, we systematically compared the model's classification performance with different state parameter settings on the IP, LongKou, Salinas, and Xuzhou datasets. The experimental results are shown in Fig. 12(a)–(d). From the results on the IP dataset, when the state parameter varies between 2 and 16, the OA of the model remains high. When state = 2, OA reaches a high level, indicating that the Mamba network has a strong capability for modeling short sequences. As the state value increases, there is no significant change in OA when state = 16. On the LongKou dataset, the impact of the state parameter on the model's performance is similarly stable. After increasing the state to 16, both OA and AA improve significantly, suggesting that in scenarios with relatively rich spectral differences and clearer boundaries, such as agricultural land, larger state values help enhance the model's ability to capture fine features and improve the synergy between local and global features. The Salinas dataset shows a different trend. Between states = 2 and 12, OA and AA remain at high levels. When the state increases to 16, both OA and κ slightly decrease although the performance remains good. This may be because the regional features in the Salinas dataset are relatively uniform, and excessively large state values tend to introduce redundant modeling, leading to slight performance fluctuations. This also suggests that in datasets with higher spectral homogeneity, blindly increasing the state parameter may not be beneficial. On the Xuzhou dataset, as the state increases from 2 to 16, OA steadily increases, and AA remains at a high level. This indicates that in complex urban environments with diverse land cover types, moderately larger state values help improve the flexibility and discriminability of sequence modeling, enhancing the model's ability to handle mixed land cover spectral information. However, further increasing the state parameter leads to excessive computational overhead, making it difficult for the computer to handle. Therefore, based on the results across all datasets, the state parameter should not be arbitrarily large, but rather needs to balance modeling ability and the risk of overfitting. Ultimately, based on the experimental performance across datasets and the model's training stability and convergence speed, this study selects state = 16 as the default configuration for the Mamba network. This setting maintains high classification accuracy across different scenarios while ensuring the model's ability to capture long-range dependencies, verifying the robustness and applicability of the proposed HyPyraMamba model.

5) Head Analysis: To further explore the impact of the number of attention heads in the PSA module on model performance, this study systematically compared the classification performance of the model with different head settings on the IP, LongKou, Salinas, and Xuzhou datasets. The experimental results are shown in Fig. 13(a)–(d). From the results on the IP dataset, as the number of heads increases from 1 to 4, both OA and κ show a slight improvement. Although the increase is not substantial, it suggests that in scenarios where the spectral band differences in hyperspectral data are significant, increasing the number of attention heads moderately can enhance the model's ability to capture diverse spectral features. However, when the number of heads exceeds 4 (e.g., 8 or 16), the performance of the IP dataset shows slight fluctuations, indicating that too many heads might introduce redundant computations and may not lead to further benefits.

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TABLE VIII
CLASSWISE ACCURACY (%) ON SPECTRALLY SIMILAR CATEGORIES ON THE IP AND XUZHOU DATASETS

Dataset	Class No.	Category	Best baseline (Method)	Best baseline Acc. (%)	HyPyraMamba Acc. (%)	Gain
IP	10	Soybean-notill	SQSFormer	90.33	90.97	+0.64
IP	11	Soybean-minitill	SSMamba	83.56	90.18	+6.62
Xuzhou	1	Bareland-1	SQSFormer	92.63	92.71	+0.08
Xuzhou	2	Bareland-2	SSMamba	99.25	99.11	-0.14
Xuzhou	3	Crops-1	MambaHS	99.44	99.80	+0.36
Xuzhou	4	Crops-2	SQSFormer	96.46	97.76	+1.30

On the LongKou dataset, as the number of heads increases from 1 to 4, OA and AA also increase, suggesting that the multihead attention mechanism helps capture spectral detail differences in larger regions, such as agricultural land. When the number of heads further increases to 8 or 16, performance changes are no longer significant, and even a slight decrease is observed in some metrics, indicating that too many heads do not necessarily bring advantages. The results on the Salinas dataset show that when head = 2, OA reaches the highest value, and AA is also relatively high. In contrast, when head = 4, OA slightly decreases, which may be related to the relatively uniform distribution of crop types and small spectral differences in the Salinas dataset, where too many attention heads might introduce overfitting risks. However, head = 4 still maintains relatively high overall performance, indicating that the model performs stably with different numbers of heads. On the Xuzhou dataset, OA reaches its highest value when head = 4, outperforming other settings, suggesting that for complex urban environments, moderately increasing the number of heads helps capture detailed information and enhances the ability to distinguish categories. However, when the number of heads exceeds 4, performance begins to decline, indicating that in urban environments, spectral–spatial features do not always require excessive attention channels for modeling, and a moderate number is sufficient. Overall, the results across the four datasets show that increasing the number of heads within a certain range indeed helps improve the model's performance, especially in scenarios with complex spectra or diverse land cover types, such as the IP and LongKou datasets. However, too many heads not only increase computational costs but may also introduce redundant information, leading to performance fluctuations. Therefore, considering model performance, training stability, and computational efficiency, this study ultimately selects head = 4 as the default setting for the PSA module. This setting achieves superior classification accuracy and good generalization ability across the four datasets, confirming the rationality and practicality of this parameter choice.

I. Classwise Analysis on Spectrally Similar Categories

To further verify the advantage of HyPyraMamba in distinguishing spectrally similar categories, we conduct a classwise comparison on several representative confusing classes, as shown in Table VIII. On the IP dataset, we focus on two soybean classes, Soybean-notill (No. 10) and Soybean-mintill (No. 11), which correspond to the same crop under different tillage conditions and thus exhibit highly similar spectral signatures. HyPyraMamba improves the accuracy on Soybean-notill from 90.33% (SQSFormer, which produced the highest accuracy on this class) to 90.97% (+0.64% points), and more importantly, sharply boosts the accuracy on Soybean-mintill from 83.56% (SSMamba) to 90.18%. On the Xuzhou dataset, we consider Bareland-1/2 (No. 1 and 2) and Crops-1/2 (No. 3 and 4), which are different surface-condition variants of bareland and agricultural fields. HyPyraMamba attains comparable performance to the best baselines on Bareland-1/2 (92.71% versus 92.63% on Bareland-1% and 99.11% versus 99.25% on Bareland-2), while improving the accuracies on Crops-1 and Crops-2: from 99.44% (MambaHSI) to 99.80% (+0.36% points) and from 96.46% (SQSFormer) to 97.76%, respectively. These classwise results demonstrate that HyPyraMamba effectively reduces mutual confusion between spectrally similar land-cover categories, providing concrete evidence for its superior discrimination ability on such challenging classes.

V. CONCLUSION

This article proposes a deep learning framework, HyPyraMamba, for HSI classification tasks, which integrates PSA, multiscale spatial–spectral fusion, and Mamba efficient sequence modeling. By introducing the PSA module, the model's ability to capture spectral features at different scales is effectively enhanced. Additionally, by combining Mamba submodules in both the spatial and spectral dimensions, the model's ability to express both local details and global context information is improved. Furthermore, the AEDC module enhances the model's adaptive handling of spatial–spectral features, addressing the issue of traditional convolutions' insufficient representation of feature diversity. Comparative experiments on four typical hyperspectral datasets (IP, LongKou, Salinas, and Xuzhou) demonstrate the effectiveness of the proposed model and its reasonable accuracy–efficiency tradeoff.

HyPyraMamba achieves higher classification accuracy than baseline methods on most datasets, and its visual results show clearer class boundaries and less classification noise, demonstrating its good adaptability to small samples and high-dimensional features in complex scenarios. Ablation experiments further validate the effectiveness of key modules. Spe_M and Spa_M play crucial roles in fine-grained spectral feature extraction and maintaining spatial structure in different scenarios. The AEDC module significantly enhances the model's flexibility and local modeling ability. The PSA module plays a key role in multiscale information capture, while the spatial–spectral fusion strategy effectively enhances the complementarity between different features, improving the overall robustness of the model. A systematic analysis of

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various experimental hyperparameters further highlights the robustness and applicability of HyPyraMamba. The experiments show that choosing a PCA dimensionality reduction of 30 strikes a balance between accuracy and computational efficiency across different datasets. An lr of 0.0003 achieves optimal performance while ensuring convergence speed. With 30 training samples, the model achieves high classification accuracy, proving its good generalization ability in small sample scenarios. Additionally, the state parameter in the Mamba module achieves good results at 16, while the PSA module performs best with head = 4, balancing accuracy and computational complexity.

Overall, HyPyraMamba demonstrates high accuracy, small performance fluctuations, and clear classification maps across different datasets and experimental configurations, further demonstrating the model's advancement and application potential in HSI classification. Although the current design is not an extremely lightweight architecture, it already achieves a favorable balance between classification accuracy and computational cost compared with high-performance baselines. Therefore, future research will further explore the lightweight design of the model and its scalability in more practical application scenarios, continuously improving the intelligent interpretation of hyperspectral remote sensing data.

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